DELPHI DIGITAL
RESEARCH REPORTS

Stablecoin Growth Propels LUNA Forward

L1 / L2 · MAR 4TH, 2021



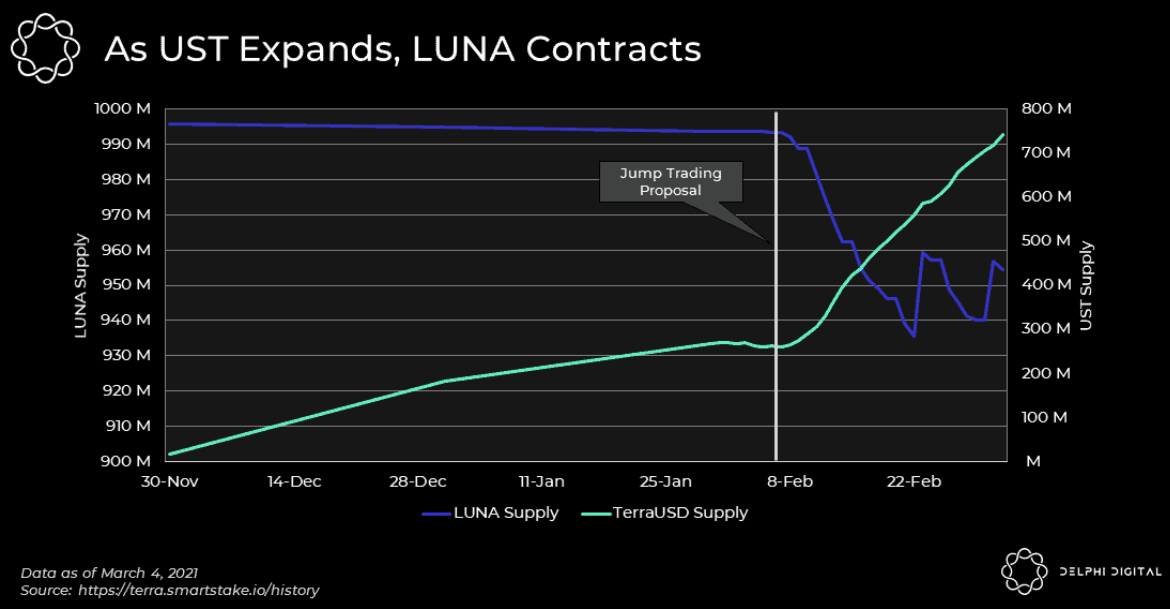
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Disclosure: Delphi Ventures and members of our team hold positions in LUNA. This statement is intended to disclose any conflict of interest and should not be misconstrued as a recommendation to purchase LUNA. This is not investment advice.



This is a follow-up post to our initial Terra report from February 2nd.

External and Internal Shifts In Wealth

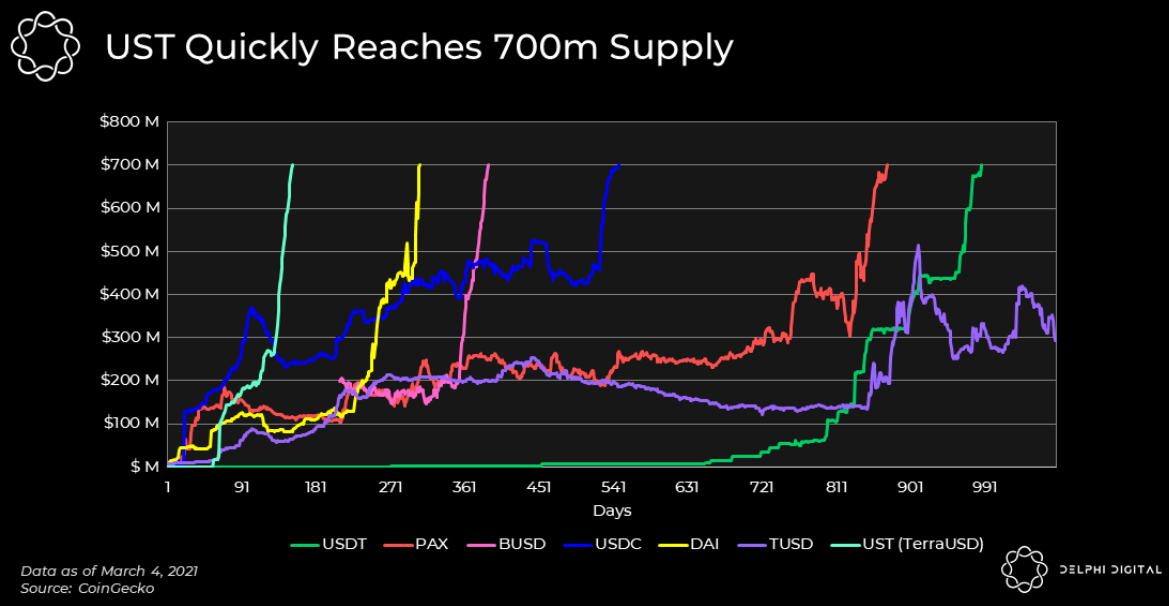
Money is purely a nominal asset. When new money is minted (i.e. printed) or burnt, wealth is not necessarily created or destroyed. Supply changes in money only make one richer or poorer if the new distribution ends up being different than the original. Even in this case, the shift in wealth is **completely internal** to the owners of that particular currency. **The total wealth of currency holders does not change, only the wealth of each holder, relative to one another, does.** True shifts in wealth only happen due to changes in demand. It's evident that Terra's team understands this principle given their focus on sourcing fundamental demand for their family of stablecoins, both from within and outside crypto.

Capital Efficient & Crypto Native Stablecoin

Compared to the existing alternatives illustrated in the chart below, Terra's stablecoins do not rely on fiat backing or over collateralization to maintain their price peg or grow supply. The demand to supply Terra stablecoins is completely aligned with the demand to use it and therefore the Terra economy is very capital efficient. The advantage of Terra, to match any market demand without constraints, became evident over the past five months where we witnessed a

rapid increase in demand for TerraUSD, primarily driven by Mirror's recent launch.

The record breaking growth of TerraUSD (UST) can be seen in the chart below. Here we compare how fast various stablecoins reached 700 million USD market cap; the current market cap of TerraUSD at the time of writing. As can be noticed, TerraUSD had a very impressive rally reaching $700m mc in only ~152 days, outpacing all existing popular alternatives. We should note, however, that many of these stablecoins launched at a time when the market was far less mature and stablecoin supplies, in general, were just a fraction of what they are today.



Seigniorage Re-Designed

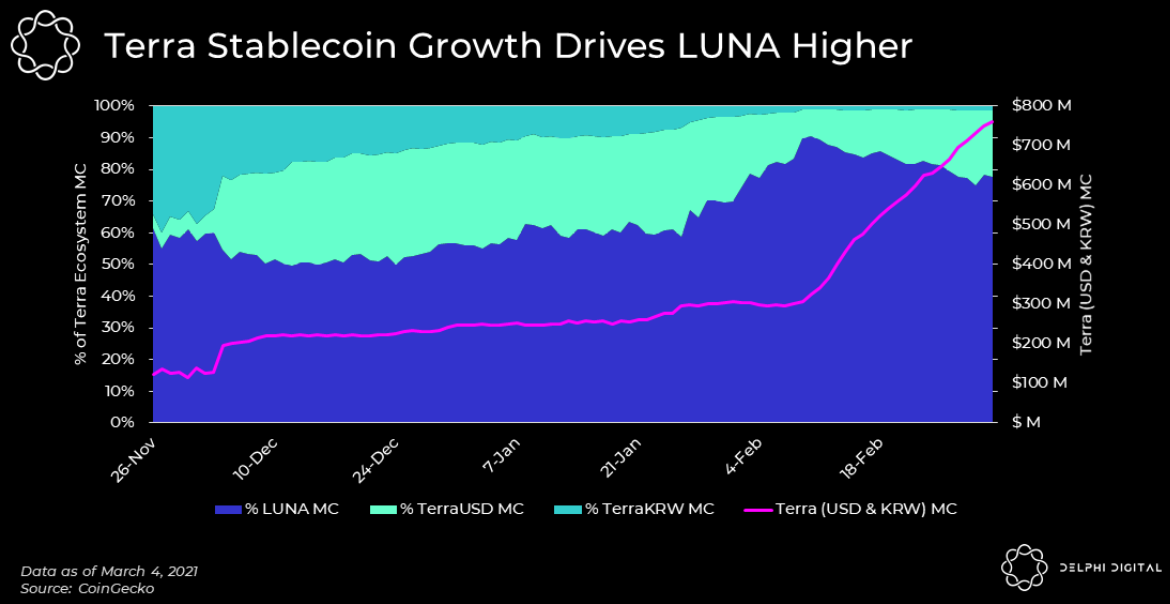
What makes Terra unique is that it has multiple fiat-pegged, algorithmic stablecoins (TerraUSD, TerraKRW, etc.) incorporated natively into the network's economic design. LUNA, Terra's native token, is used to both stake the network and collateralize the algorithmic stablecoins. Miners in the Terra blockchain are the equivalent of bondholders in the traditional seigniorage model. They stake LUNA to secure the network and earn transaction fees from Terra transfers, as well as seigniorage rewards.

In brief, here is how the on-chain swap stability mechanism works during stages of expansion and contractions:

Expansion: If demand causes the market price of UST >$1, arbitrageurs can burn

$1 of LUNA to mint 1 UST, which can then be sold spot at the higher price for a riskless profit. Spot selling drives UST price back to its $1 peg, while minting expands UST supply. This process reduces the LUNA supply which is value-accretive to the network's stakers.

Contraction: If demand causes the market price of UST <$1, arbitrageurs can buy UST spot and redeem it at par value (i.e. $1) for newly minted LUNA. Spot buying drives UST price back to its $1 peg, while redemptions contract the UST supply. This process increases the LUNA supply which is value-dilutive to the network's stakers.

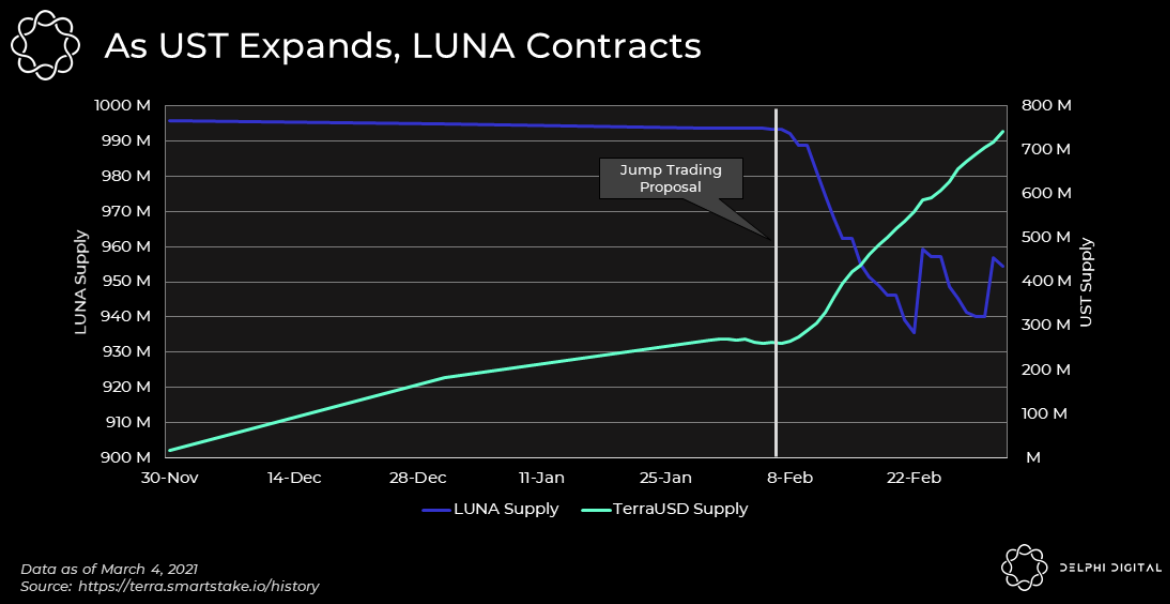


Expansion's happen due to high demand and are usually the easy part. The real risk is during contractions. It's important that the protocol does not mint more LUNA than the market can comfortably absorb, because any LUNA that gets minted dilutes miners who have LUNA staked. To compensate for dilution, the system rewards miners with higher unit rewards by handing out tx fees and burning a higher portion of seigniorage. However, if stablecoin demand falters, the protocol could keep minting new LUNA to buy back Terra stablecoins, depressing the LUNA price. If declining mining rewards can no longer compensate miners against dilution, a vicious downward cycle can start.

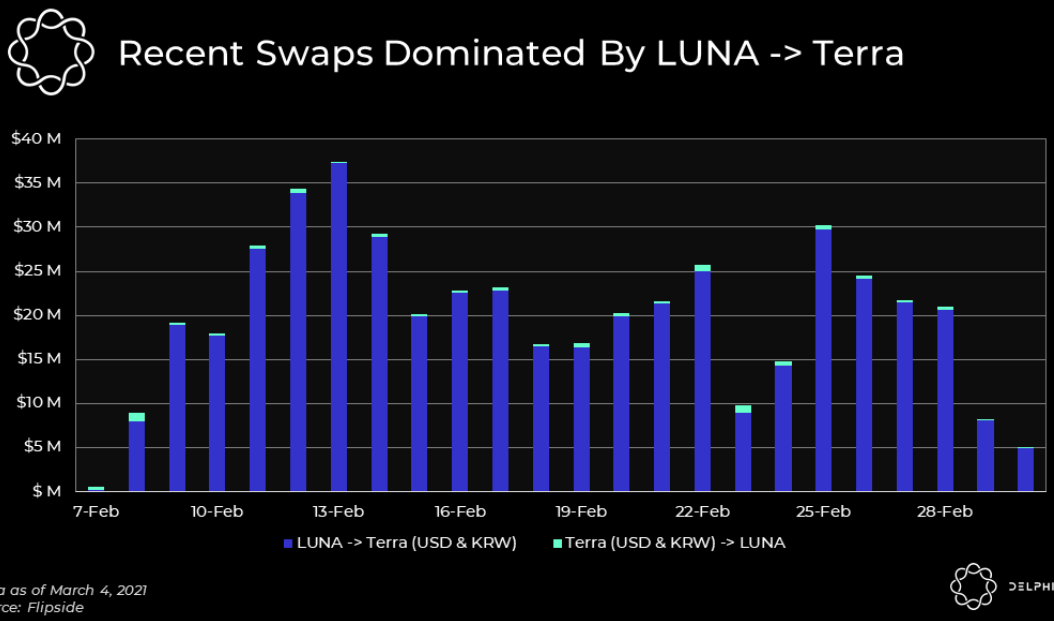
The best way to protect against this is to source **sustainable stablecoin demand through increased utility**. Currently, the main demand driver is a relatively new synthetics platform, Mirror, but there should be other applications coming online

in the near-term. Projects like Anchor and Alice will soon launch on Terra further contributing to this demand. It's important to note that Terra also sources demand from outside the crypto-native community. This is extremely important because, as seen in the past, crypto can be subject to sudden demand shocks and heightened volatility. This is why Terra's long-term plan is to continue increasing stablecoin usage in e-commerce. **A constant source of relatively inelastic demand is key for Terra's sustainable growth.**

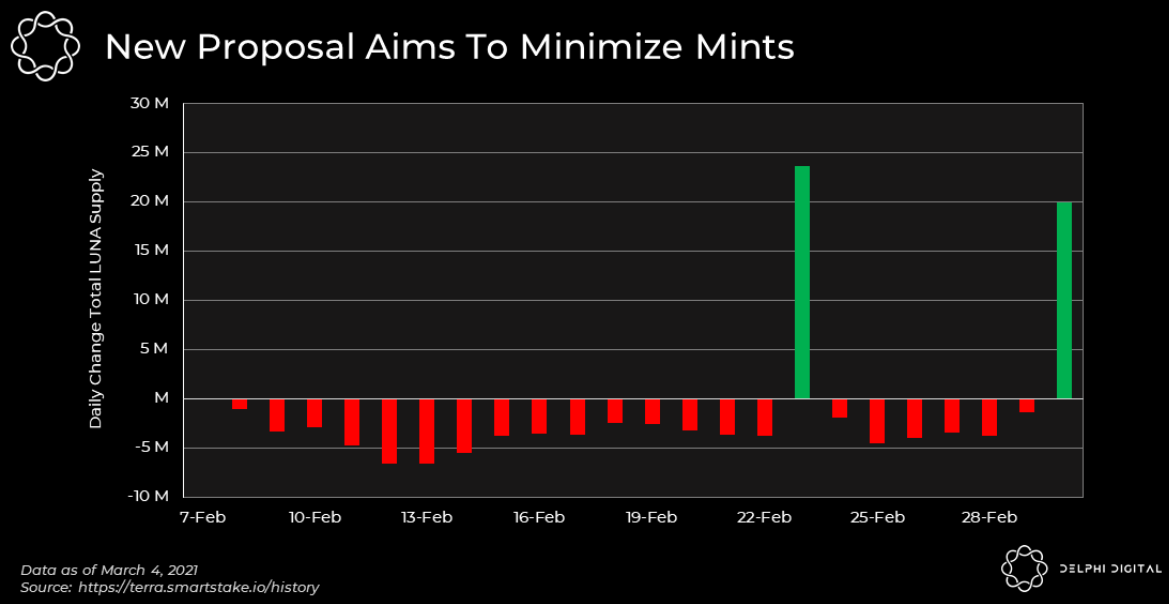
Improving On-Chain Swap Liquidity



Until recently, a large portion of stablecoin growth was facilitated by the Terra Labs Foundation via direct sales from its reserves, instead through on-chain swaps. Although not ideal, this was necessary at the time due to the high slippage costs of their on-chain swaps caused by low liquidity. This, however, changed dramatically after a governance proposal, put forward by Jump Trading, was implemented on Feb 7. This proposal was focused on improving on-chain liquidity. **Since then, ~470 M USD worth of LUNA has been swapped to mint Terra, burning 4.3% of LUNA's total supply over the span of a few weeks.** TerraUSD supply rose above 700 million and the protocol accumulated a lot of seigniorage in a very short time period.



As the protocol began to accumulate significant amounts of seigniorage, we witnessed large payouts to stakers and the treasury at the end of the last 2 epochs, which can be noticed in the chart below. As per the governance parameters on Feb 23, off of the ~59M LUNA seigniorage rewards that accumulated since Feb7, ~12M moved into Treasury and another ~12M to stakers while the rest was burned (destroyed forever). Same thing happened on March 2; this time with a new distribution proportion of 47.5/47.5/5 to Treasury, Stakers and burn address respectively.



Large LUNA mints on February 23 and March 3 are the result of governance parameters which mint a portion of the total absorbed seigniorage back into

existence at the end of an epoch in order to route them to Treasury and Stakers. However, this is likely to change very soon as the community thinks the Treasury is already over-funded and stakers income can be kept at attractive levels even without seigniorage gains. On this point, a community proposal that is soon expected to pass **suggests burning all of seigniorage gains moving forward**, preventing the large LUNA issuances at the end of epochs. This implies that **$1 worth of LUNA must be destroyed to mint a single TerraUSD.**

With the introduction of the new governance parameters and upcoming DeFi app launches, LUNA may be poised to continue its deflationary trend in the foreseeable future. With that being said, onlookers need to watch for contractions in Terra stablecoin demand because it is LUNA that will absorb the knock-on effects of it.